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comp 009 - OOP

BSIT 2-4

notes

MIDTERM ASSIGNMENT #1 - INTRODUCTION TO JAVA

1. What is Java programming language? Discuss its features and advantages.

According to Oracle (2019), Java is created by James Gosling at Sun Microsystems in 1995. It is a high-level object-oriented language emphasizing simplicity, security, and portability. It powers everything, including apps on Android-run devices, to enterprise servers.

Its key features include an object-oriented design, supporting the 4 pillars. It could also run anywhere, in most environments, being called a "write once, run anywhere" type of programming language. As a student who started coding with C language, its similar syntax with Java also makes it a feature, and an advantage to me. It makes it easy for me to understand the basics, leaving only the feature I have to understand and learn. An advantage of Java would be its cross-platform compatibility making it easy for environments to run. In addition, Java has a vast ecosystem for web, mobile, and big data.

2. Explain the concept of platform independence in Java. How does Java achieve platform independence?

Platform independence in Java lets the code run unchanged across operating systems such as Windows, Mac, and Linux. It achieves this via compilation to intermediate bytecode executed by the JVM, which translates it to native machine code. Each platform's JVM handles the specifics, embodying the write once, run anywhere (Oracle, 2019).

3. Discuss the main components of a Java programs and their roles.

According to Eckel (2019), a basic Java program should contain the following:

1. Class - every Java program must have this. The blueprint for objects.

```
• public class main {
```

2. main method - The entry point of the program. It starts executing from here.

```
• public static void main (String [] args) {
```

3. Statements - These are the instructions inside the main method (printing, etc).

```
• System.out.println ("Hello World");
```

4. Variables and Data Types - explicit data types.

```
• int, double, char, boolean, String
```

5. Objects and methods - objects are instances of classes, and methods define behaviors

4. What are the differences between JDK, JRE, and JVM?

According to Oracle (2024), the differences of the three are as follow:

1. JVM (Java Virtual Machine) - is responsible for executing Java bytecode. It converts bytecode into machine-specific code, enabling platform independence.

2. JRE (Java Runtime Environment) - it contains JVM, core libraries, and supporting files that are needed to run Java programs. Again, runs Java applications but not to develop them.

3. JDK (Java Development kit) - It contains JRE, compilers, and development tools for debugging and documentation. Basically, used to develop and compile Java programs.

All in all, these three are different in terms of its use. one cannot function without the other, and so on. making it a stage-by-stage execution. JDK has JRE, and JRE contains JVM.

5. Explain the difference between interpreted and compiled languages. Is Java an interpreted or compiled language?

When it comes to compiled languages, these are languages compiled and translated into machine code before execution. Meaning, the entire source code is translated to be read by the machine, and can run directly on the system. On the other hand, interpreted languages translates and executes the code line by line when run. Java in itself, is both compiled and interpreted.

Since a Java source code typically has .java in its file, it is compiled into bytecode using the compiler, which is the javac. The JVM then interprets or just-in-time compiles the bytecode into machine code during execution. Therefore, making Java use a hybrid approach.